

APICULTURE:

- ❖ Beekeeping is defined as maintenance of beehives by humans.
- ❖ Honeybees are considered as important pollinators.
 - ❖ **Different species of honeybees**
 - There are four well known species of true honeybees (belonging to genus *Apis*) in the world:
 1. Rock bee, *Apis dorsata*
 2. Little bee, *A. florea*
 3. Asian bee, *A. cerana*
 4. European bee, *A. mellifera*
 - ❖ **Characteristics of four well known species of honeybees**

	<i>Apis dorsata</i>	<i>Apis florea</i>
Nesting	Open nesting. Builds single large comb (ca 1m ²) attached to branches of trees or rocks etc.	Open nesting. Builds single small comb (ca size of palm of hand) fixed to branches of bushes.
Distribution in India	Found in plains as well as hills up to 1600 meters above sea level. Highly migratory.	Found in plains up to 300 metres above sea level. Highly migratory.
Size	Biggest honeybee (16-18mm)	Smallest <i>Apis</i> bee (9-10mm)
Average honey yield per colony/year	40 kg (wild bees; cannot be domesticated)	500 g (wild bees; cannot be domesticated)
Method of Honey extraction	By squeezing (unhygienic)	By squeezing (unhygienic)
Number of cells/10cm comb (worker cells)	18-19	32-36

	<i>Apis cerana</i>	<i>Apis mellifera</i>
Nesting	Cavity nesting. Builds many parallel combs in cavities of tree trunks, hollows of rocks, poles and	Cavity nesting and similar in habits to <i>Apis cerana</i> and builds parallel combs.

	other covered places	
Distribution in India	Found throughout India	Exotic bee to India. Introduced successfully in 1962.
Size	Medium size (14-15mm)	Medium size (14-16mm)
Average honey yield per colony/year	3 kg (Hive bees; can be domesticated)	15 kg Hive bees; can be domesticated
Method of Honey extraction	By centrifugal honey extractor from the hived bees (hygienic).	By centrifugal honey extractor from the hived bees (hygienic)
Number of cells/10cm comb (worker cells)	21-25	17-19

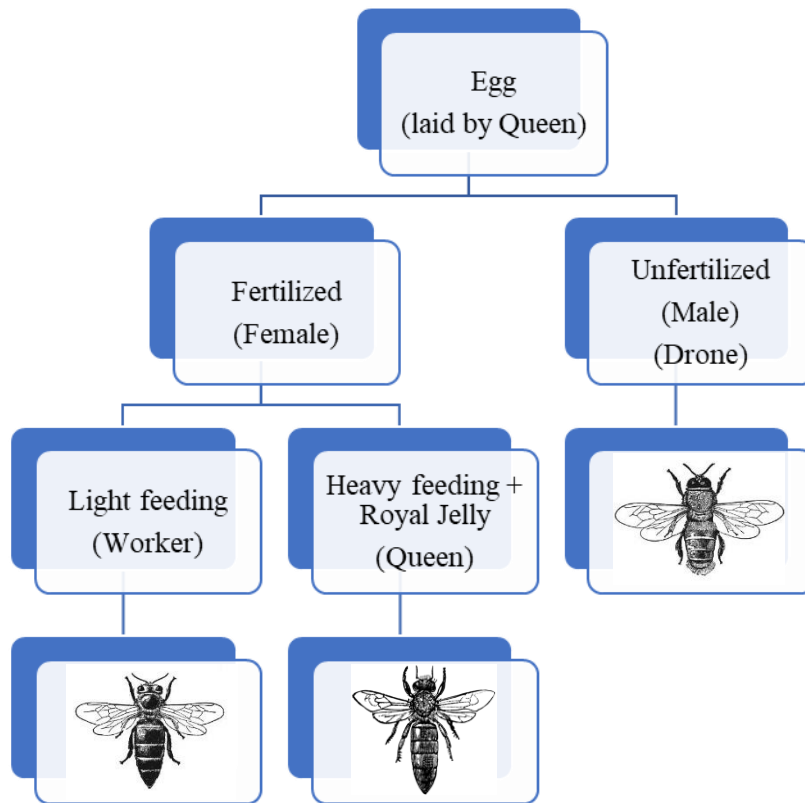
❖ **Like any insect, body of honeybee can be distinguished in to three parts:**

1. Head
2. Thorax
3. Abdomen

1. Head

- Bears a pair of geniculate antennae.
- Mouth parts of worker bees are modified for sucking and lapping.
- Inside the head there are long coiled strings of small lobes known as hypopharyngeal glands which secrete glandular food known **as royal jelly** that is fed to queen and young larvae.
- **Prothoracic legs serve as antenna cleaner.**
- On Mesothoracic legs, **bushy tarsi serve as brushes for cleaning of thorax.**
- In worker bees, smooth somewhat concave **outer surface of hind tibia is fringed with long curved hairs and forms pollen basket or corbicula.**
- In workers **egg laying apparatus (ovipositor) is modified into sting.**
- Queen uses ovipositor for **egg laying and stinging rival queen.**
- Digestive system is unique in having oesophagus with expanded **honey stomach which stores the collected nectar.**
- Sperms are stored in the queen in a sac like structure known as **spermatheca**. The stored sperms are utilized by queen throughout her lifetime as she does not go for mating once starts egg laying.

- A normal colony, during active season is composed of 3 kinds of individuals: one queen, thousands of workers (10000 to 30000 or even more) and few hundreds of drones, which vary in size. **In addition, each colony has different developmental stages viz eggs, larvae and pupae which are collectively known as brood.**



❖ Queen:

- Only one queen is found in a colony except under supersedure or swarming instinct.
- She is mother of whole colony producing workers and drones and is the only perfectly developed female member of the colony.
- Her function is to lay eggs. She does not have motherly instinct or ability to feed the brood. She is fed lavishly by a large number of nurse bees with highly nutritious food known as royal jelly.
- A good queen can lay **1500-2000** eggs per day.
- A laying queen is the **longest bee** in the colony.
- **Fertilized eggs produce workers (also queens) and unfertilized eggs produce drones.**
- A good, mated queen may work satisfactorily for 2 or more years, although queens can live eight years or longer.

- The virgin queen mates with a number of drones (5-7) within 5-10 days of emergence in the air (not inside the hive) and spermatozoa are stored in spermathecal. Stored sperms are utilized to fertilize eggs throughout her life till exhausted.

❖ **Worker:**

- Workers are imperfect females. They are unable to mate though they may start egg laying if a colony remains queen less for long period.
- The workers perform all the useful work in the colony.

❖ **Laying workers:** Under queen less conditions for a long duration, ovaries of some of the workers start developing and they can lay even eggs but since these are unfertilized, give rise to only drones.

❖ **Drone:**

- Drones do not perform any duty inside the hive.
- A colony rears and tolerates the drones only during breeding season.
- The **sole function** of a drone is to mate once which costs him his life.
- Maximum life of drone honeybee in summer is 59 days

❖ **Life cycle:**

- Queen Deposits egg at the base of cell and fastens with mucilaginous secretion.
- After 3 days egg hatches and workers provide pearly white food in which “C” shaped larva floats.
- Cell is sealed when larva is fully grown. In the sealed cell it turns into pupa from which adult emerges.
- Larva sheds skin five times during development.
- The sealed cells containing worker and drone brood and honey can be differentiated based on appearance.

Caste	Egg period (days)		Larval Stage(days)		Pupal Stage (days)		Total (days)	
	A. cerana	A. mellifera	A. cerana	A. mellifera	A. cerana	A. mellifera	A. cerana	A. mellifera
Queen	3	3	5	5	7-8	8	15-16	16
Worker	3	3	4-5	5	11-12	12-13	18-20	21
Drone	3	3	7	7	14	14	24	24

- ❖ **Trophallaxis** is food transmission (exchange of food) which is common between workers and also from workers to queen and drones. It is a sort of communication regarding availability of food and water and also a medium for transfer of pheromone.